6. NOR operator is commutative bc

$(a+b)' = (b+a)'$

7.

$a+c'+0 = a+c'$

$a+c' = a+c'$

8.

$1 \quad 001 = x'y'z$

$2 \quad 010 = x'yz'$

$6 \quad 110 = xyz'$

$7 \quad 111 = xyz$

$(x'y'z + x'yz' + xyz' + xyz$

XOR

$x'(y'z + yz') + xy(z'+z)$

$x'(y \oplus z) + xy$

9.

$1 \quad 001 \quad x+y+z'$ Cannonical form = product of sums

$5 \quad 101 \quad x+y+z'$

$7 \quad 111 \quad x'+y+z'$

$0 \quad 000 \quad x+y+z$

$(x+y+z')(x'+y+z')$
$(x'+y'+z')(x+y+z)$

10.

10) $abcd + \cancel{abd} + abd + a'bc + ① + a'bd$
$+ b'cd + cb'da\cancel{b} + b'd + c'ab + c'b'bd$

$ab'd + a'bc + a'bd + b'dc + b'd + c'ab + c'bd$

$bd(a' + a) + c(a'b + b'd) + b'd + c'ab$
$bd$

$= bd + b'd + ca'b + c'ab$

$(ba + bc)(a' + c')$

$= b(a + c) + d(b + c)$

$bac' + bca' + db + dc$

Equation is valid

1.) 7 4 2 9
0111 0100 0010 1001

2) 0011 0101
3 5

3.) 2421 has complementing. 9's complement found by flipping bit

4) Parity bit transmitted is even (0)
(a)

b) Error is detected bc the # of 1s would be inconsistent with parity bit and retransmission would be requested

6.) $\frac{7!}{2!(7-2)!} = \frac{42}{2} = 21$

Odd # of errors change parity
11 errors change parity